

Labwork 5: Gaussian Blur Convolution

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October 2, 2026

1 Objective

The goal of this labwork is to compare an image blur program on a CPU and a CUDA GPU. Three versions were tested: CPU, GPU with shared memory, and GPU without shared memory.

2 Method

The input image size was 6000×2962 , or 17,772,000 pixels. The GPU versions were tested with three block sizes: 8×8 , 16×16 , and 32×32 . The measured execution times were then compared.

3 Results

Implementation	Block size	Time (s)
CPU	N/A	187.7
GPU with shared memory	8×8	0.064995
GPU with shared memory	16×16	0.061779
GPU with shared memory	32×32	0.066184
GPU without shared memory	8×8	0.063658
GPU without shared memory	16×16	0.061930
GPU without shared memory	32×32	0.061409

Table 1: Execution time of the two GPU versions.

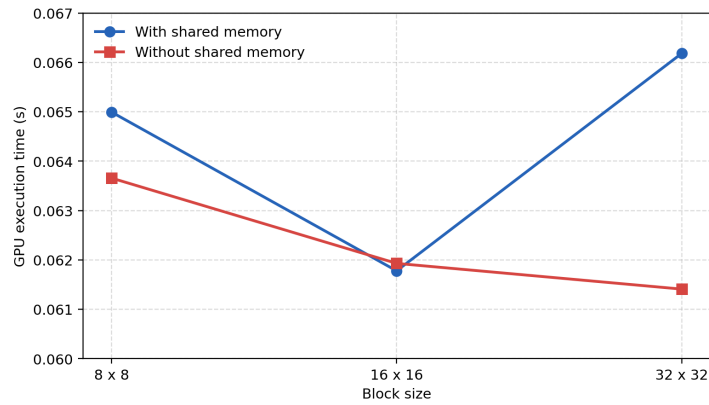


Figure 1: Block size versus execution time for the two GPU versions.

The fastest shared-memory result was 0.061779 seconds with a 16×16 block. The fastest result overall was 0.061409 seconds without shared memory, using a 32×32 block.

4 Discussion

Changing block size did not significantly change the GPU time. With shared memory, the 16×16 block was the best choice. Without shared memory, performance improved as the block size increased. In this test, shared memory did not give a clear benefit.